

Table 2: Table Summary

No.	Authors / Year	Title / Source	Mapped RQ(s)	Objectives / Focus	Methods / Models Used	Key Findings / Contributions	Relevance to the Project (Usefulness or Gap)
S1	Jerbi, H.; Vincy, V.G.A.; Ben Aoun, S.; Abbassi, R.; Kchaou, M. (2025)	<i>Optimizing waste management in smart cities: An IoT-based approach using Dynamic Bald Eagle Search Optimization (DBESO) and machine learning (Scopus)</i>	RQ1, RQ2, RQ3, RQ5	Propose an IoT- and ML-based smart bin monitoring and waste management system.	IoT sensors + Dynamic Bald Eagle Search Optimization (DBESO) + Kernel Soft Extreme Learning Machine (KSELM).	Achieved up to 96.42 % accuracy in bin status prediction and waste forecasting; improved precision (94.9–96.3 %) across iterations.	Shows IoT + ML forecasting for route optimization; lacks continuous IoT sensing and full Digital Twin integration.
S2	Taimur Bakhshi and Muhammad Ahmed, 2018	“IoT-Enabled Smart City Waste Management using Machine Learning Analytics” <i>Proceedings of the 2018 International Conference on Energy Conservation and Efficiency (ICECE), Lahore, Pakistan, pp. 70–76.</i>	RQ1	Propose an IoT-based system for smart waste monitoring and route optimization.	Raspberry Pi-based IoT architecture with ultrasonic sensors for bin fill-level monitoring. Machine learning analytics were applied for regression and classification of fill-level data, but the specific ML model names were not specified by the authors. Dynamic route optimization was performed based on predicted fill levels	Improved fuel efficiency by 46 % and reduced collection time by 18 % during trials.	Shows IoT + ML benefits; lacks continuous sensing and Digital Twin integration.
S3	Morais, C., Cardoso, S., & Oliveira, J. F. (2024)	<i>A Data-Driven Optimization Approach to Plan Smart Waste Collection Operations (Scopus)</i>	RQ1, RQ2, RQ3, RQ5	To reduce uncertainty in waste collection routes using real-time bin data.	IoT sensors, HMM forecasting, dynamic reverse inventory routing, GAMS + CPLEX optimization.	Improved efficiency and service levels; reduced overflow; solved real cases under 30 mins.	-The project aligns with AI forecasting and dynamic route optimization objectives lacks continuous IoT and Digital Twin integration.

S4	Lunawat, A.; Rastogi, V.; Shrivastava, A.; Singh, M.; Ayush; Harikrishnan, R. (2023)	<i>A Smart Framework for IoT Based Garbage Monitoring System</i> — Symbiosis Institute of Technology report - IEEE	RQ1	Develop an IoT-based system to prevent bin overflow and improve waste collection.	IoT setup with Raspberry Pi, ultrasonic sensors (HC-SR04), LCD, cloud (ThingSpeak).	Real-time monitoring alerts municipal services when bins are full, reducing human error and overflow.	Useful for IoT sensing; lacks ML forecasting and Digital Twin-based route optimization.
S5	S. Sharmila Devi; S. Ajae Dhanapal; P. Gokul; M. Gokul; S. Rathish; R. Ranjith Kumar (2025)	<i>Automatic Garbage Collector using Sensors and Internet of Things</i> — Proc. ICTMIM-2025 (IEEE)	RQ1	Develop an IoT-based automated waste collection system to minimize human effort and improve efficiency.	IoT setup with ESP32, Arduino, ultrasonic sensors, servo motor, electromagnets, line follower robot, and MQTT communication.	Achieved fully automated waste collection with real-time monitoring, obstacle detection, and reduced labor dependency.	Useful for automation and IoT integration; lacks ML forecasting, route optimization, and Digital Twin simulation for scalability.
S6	Billah, M. M.; Sultana, A.; Rahaman, M. M. (2025)	<i>An Optimized IoT-Based Real-Time Categorized Smart Waste Management System</i> — IEEE ECCE 2025	RQ1, RQ3	The system aims to improve efficiency by monitoring waste bin fill levels, categorizing waste, and optimizing routes for waste collection vehicles	IoT sensors, GPS routing, CRN (CNN) model for waste classification, Google OR-Tools for route optimization.	Achieved 95% accuracy in waste categorization, reduced fuel consumption and costs, and improved waste collection efficiency.	Relevant for IoT and route optimization; lacks Digital Twin integration and continuous sensing for predictive and adaptive optimization.
S7	Tran Anh Khoa, Cao Hoang Phuc, Pham Duc Lam, Le Mai Bao Nhu, Nguyen Minh Trong et al. (2020)	<i>Waste Management System Using IoT-Based Machine Learning in University, Wireless Communications and Mobile Computing (Scopus)</i>	RQ1	Develop a low-cost IoT and ML-based waste management system to optimize collection in a university setting.	IoT sensors with LoRa, Logistic Regression for prediction, Dijkstra's algorithm for route optimization, graph theory, mobile app interface.	Achieved 0.99 AUC accuracy, optimized routes, reduced labor costs, and improved collection efficiency.	Highly relevant for ML-based route optimization; lacks Digital Twin integration and large-scale deployment validation.
S8	Alexopoulos, K., Catti, P., Kanellopoulos, G., Nikolakis, N., Blatsiotis, A., (2023). <i>Deep Learning for Estimating the Fill-Level of Industrial</i>		RQ1	Develop a deep learning-based computer vision system to estimate	VGG16 CNN, ImageNet pre-trained model, IIoT	Achieved 77.5–95% accuracy across test sets; validated on real industrial data	Relevant for vision-based fill-level forecasting; lacks real-time IoT sensing, route

S9	Christodouloupoulos, K., Kaimenopoulos, A., & Ziata, E.	<i>Waste Containers of Metal Scrap: A Case Study of a Copper Tube Plant. Applied Sciences</i> , 13(4), 2575.		industrial waste container fill levels.	integration, low-cost hardware setup.	from a copper tube plant.	optimization, and Digital Twin integration.
	Paavan Lakshmana Chowdary S.; Sai Teja G.; Naga Mahesh K. (2019)	<i>An IoT based Smart Garbage Alert System — Proc. ICOEI 2019 (IEEE)</i>	RQ1	Design an IoT-based smart garbage alert system to prevent overflow and improve waste management efficiency.	NodeMCU ESP8266 microcontroller, sensor network for level detection, cloud data transmission via IoT.	Automated alerts when bins are full; improved monitoring efficiency; scalable for crowded areas.	Relevant for IoT-based sensing and alerting; lacks ML forecasting, route optimization, and Digital Twin integration.
S10	Md. A. Rahman, S. W. Tan, A. T. Asyhari, I. F. Kurniawan, M. J. F. Alenazi, and M. Uddin (2024)	<i>IoT-Enabled Intelligent Garbage Management System for Smart City: A Fairness Perspective. IEEE Access.</i>	RQ1	Develop a fairness-based IoT-enabled waste collection framework using LoRa communication to improve routing efficiency and resource utilization.	LoRa-based IoT network for data exchange, Capacitated Vehicle Routing Problem (CVRP) formulation with fairness objective function.	Achieved balanced workload among trucks and reduced total distance; demonstrated reliability and low-cost deployment with LoRa.	Relevant for IoT-enabled route optimization and fairness modeling; lacks Digital Twin integration and ML-based dynamic forecasting.
S11	V. Chithra, A. Deepika, S. Karthikeyan, R. Kiruthiga, M. Nithya (2025)	<i>Decentralized and Intelligent Waste Management System Using Blockchain and AI — IEEE</i>	RQ1, RQ2, RQ3, RQ5	Develop a decentralized intelligent waste management system integrating Blockchain, AI, and IoT to enhance efficiency, security, and transparency.	Deep Neural Network (DNN) with Adam Optimizer, IoT smart sensors, Private Ethereum Blockchain (PoA), Smart Contracts for automation.	Achieved 92.5% collection efficiency, 96.2% prediction accuracy, and 94.8% trust score; improved automation and incentive fairness.	Highly relevant for AI forecasting and IoT sensing integration; lacks Digital Twin-based simulation and real-time dynamic optimization.
S12	Chavan, S.; Patil, U.; Koshy, S. S. S.; Srikanth, S. V. (2021)	<i>Garbage Zero (Garb0): An IoT Framework for Effective Garbage Management in</i>	RQ1	Develop and evaluate Garb0, a low-cost, low-power IoT-based waste	IoT sensors with LPWAN LoRa, ARM Cortex-M0+ microcontroller, CooCox CoOS software, Google	Achieved 1-year battery life, reliable real-time monitoring, and optimized	Relevant for IoT monitoring and route optimization; lacks ML forecasting and Digital Twin integration for

		<i>Smart Cities — IEEE</i>		monitoring system for smart cities.	API-based route optimization.	collection routes through low-power design.	predictive analytics and visualization.
S13	Namoun, A.; Tufail, A.; Khan, M.Y.; Alrehaili, A.; Syed, T.A.; BenRhouma, O. (2022)	<i>Solid Waste Generation and Disposal Using Machine Learning Approaches: A Survey of Solutions and Challenges - scopus</i>	RQ2, RQ3	Review and analyze machine learning approaches for optimizing waste generation and disposal processes in smart cities.	Reviews ML models : ANN, CNN, GBRT, Decision Trees, IoT-based systems, and Blockchain frameworks.	CNN achieved highest accuracy (96%) in waste classification; main challenges include lack of real-time data, benchmarking, and public datasets.	Highly relevant for ML forecasting insights; highlights data scarcity and absence of Digital Twin or route optimization—gaps the project aims to address.
S14	M. Pollak, M. Tovarek, D. Ourednicek, M. Kulhanek, P. Chmelar, and J. Faigl (2024)	<i>Optimized Operation Management with Predicted Filling Levels of the Litter Bins for a Fleet of Autonomous Urban Service Robots — IEEE</i>	RQ2, RQ3, RQ5	Optimize autonomous waste collection through ML-based fill-level prediction and route planning.	XGBoost Classifier, Simulated Rebalancing with Knapsack algorithm, compared to Nearest Neighbor baseline.	Achieved 82% prediction accuracy; reduced operational time by 26–49% and energy use by 31–34%.	Highly relevant for ML forecasting and route optimization; lacks IoT real-time sensing and Digital Twin validation.
S15	Ch. Anuradha, M. Pothuraju, N. Manam, C. Chigurupati (2024)	<i>IoT-Based Bin Level Monitoring for Efficient Waste Management — IEEE</i>	RQ2, RQ3, RQ5	Automate waste collection using IoT to improve efficiency and sustainability.	IoT sensors with ultrasonic detection, Dijkstra's algorithm for route optimization, Linear Regression for fill-level prediction.	Reduced costs and unnecessary trips; achieved real-time monitoring and optimized routing for efficient collection.	Relevant for IoT-based monitoring and route optimization; lacks Digital Twin integration and advanced ML forecasting for dynamic adaptation.
S16	D. Thayalnayaki; J. Santhosh; S. J. Princess Rosaline; P. Latha; V. A. Shanmugavelu; Tamilmani A. (2023)	<i>Machine Learning and IoT-Based Solid Waste Management: Revolutionizing Efficiency and Sustainability for a</i>	RQ2, RQ3	Combine IoT sensing with ML forecasting to predict bin fill levels and support smarter collection planning.	IoT sensors (ultrasonic, infrared, weight), ML models: ANN, LSTM, Random Forest for fill level prediction.	Achieved high prediction accuracy (ANN 98.3%, LSTM 95.5%, RF 93.55%); reduced unnecessary collections and operational costs.	Highly relevant for ML forecasting and IoT sensing; lacks route optimization and Digital Twin integration for full system automation.

S17		Clean Environment — (IEEE)					
	S. Tiwari, M. Pandey, R. Sharma, P. Gupta, A. Yadav (2025)	Smart Waste Management System using AR and IoT with AI — IEEE	RQ2	Develop a smart waste management system integrating IoT, AR, and AI to improve urban sustainability and operational efficiency.	IoT sensors for real-time monitoring, YOLO + CNN + KNN algorithm for waste classification, AI models for route optimization, AR interface for visualization, Firebase for data management.	Improved recycling efficiency, reduced fuel use, and enhanced public awareness through AR visualization and real-time categorization.	Highly relevant for IoT and AI-based forecasting and route optimization; lacks Digital Twin integration for simulation and scalability testing.
	Kalaiselvan, C.; Sowmya, G.; Nageswararao, K. (2025)	An Automated Smart Waste Management Systems using Support Vector Machine — Proc. ICICV 2025 (IEEE)	RQ2	to classify bins and improve efficiency. using ML to support smarter collection decisions.	SVM for bin classification, data preprocessing, and feature extraction.	Achieved 98.2% accuracy and 90.3% F1-score; reduced labor and fuel costs while enhancing sustainability.	Relevant for ML-based automation and efficiency; lacks IoT real-time sensing and Digital Twin integration for dynamic route optimization.
	Rodríguez López, J.C., Rodríguez- García, M.I., Moscoso López, J.A., Ruíz Aguilar, J.J., Alcántara Pérez, J.M., & Turias Domínguez, I.J. (2021)	Prediction of Container Filling for the Selective Waste Collection in Algeciras (Spain) (Scopus)	RQ2, RQ3	Develop an intelligent system to predict cardboard container fill levels for efficient and sustainable waste collection.	Machine learning models (ANN, MLP, multiple and nonlinear regression) using historical fill-level data with cross- validation.	Achieved 26% higher sensitivity and 67% higher specificity; reduced costs, fuel use, and emissions while improving service quality.	Highly relevant for AI- based fill-level forecasting and optimization; lacks IoT real-time sensing and Digital Twin integration for automated route simulation.
S20	K. Yesu Mariyamma, G. RamaChandra Rao (2024)	An Automated Smart Waste Management System using DT — IEEE	RQ2, RQ3	To design an IoT- based automated waste management system that predicts bin status using machine learning to enhance monitoring efficiency.	Machine learning models — Decision Tree and Logistic Regression (Decision Tree performs best).	Achieved high precision, recall, and F1-score in predicting bin status; reduced unnecessary trips and resource waste.	Relevant for ML-based fill-level prediction; lacks IoT real-time sensing and Digital Twin integration for route optimization and dynamic monitoring.

S21	Velumani Thiyagarajan, L. Jaya Singh Dhass, Vensila C., Ben M. Jebin, Seethalakshmy Anantharaman (2025)	<i>Intelligent Waste Management System using Digital Twin Technology</i> — IEEE	RQ2, RQ3	Aimed to develop an intelligent waste management system using machine learning (decision trees) for predicting bin fullness, addressing environmental and health impacts. However, the term <i>digital twin</i> was used conceptually, not implemented in practice.	IoT-enabled smart bins with cloud and adaptive systems; applied decision tree, logistic regression, and deep learning; used federated learning and blockchain; <i>digital twin</i> only mentioned conceptually.	Demonstrated high precision, recall, and F1-score in predicting dustbin status using decision tree models, outperforming logistic regression and achieving accurate fill-level prediction.	Supports AI-driven fill-level forecasting using decision trees but lacks real-time IoT sensing and digital-twin-based route optimization. The proposed project addresses these gaps through integrated IoT data collection and dynamic digital-twin routing.
S22	Linard Barth, Lukas Schweiger, Rodolfo Benedech, Matthias Ehrat (2023)	<i>From Data to Value in Smart Waste Management: Optimizing Solid Waste Collection with a Digital Twin-Based Decision Support System</i> , <i>Decision Analytics Journal</i> (Elsevier)	RQ2, RQ3, RQ4, RQ5	Aimed to optimize solid waste collection through a digital twin-based decision support system that models the trade-off between cost savings and service quality using sensor-integrated data.	Used a mixed simulation approach (agent-based and discrete event) in AnyLogic with GIS (OpenStreetMap) and OptQuest optimization. Integrated IoT time-of-flight sensors and an industry-linked digital process twin for decision support.	Demonstrated that digital twin-based optimization can cut costs by up to 73% while maintaining service quality. Introduced a new KPI for service quality, provided municipal benchmarks, and validated the model's ability to balance cost and environmental impact.	Provides a digital process twin for cost-quality optimization but relies on static routing and lacks AI-based fill-level forecasting. The proposed project overcomes these gaps through predictive ML models and dynamic route optimization within a real-time digital twin.
S23	Dung D. Vu, Georges Kaddoum (2017)	<i>A Waste City Management System for Smart Cities Applications</i> — IEEE	RQ2, RQ3, RQ5	Aimed to enhance smart city waste management by predicting bin status, classifying bin locations, and optimizing collection routes for cleaner	Used K-means clustering (with Elbow method) for bin grouping, logistic regression for fill-status prediction, and a Genetic Algorithm for route optimization, applied on an	Enabled automatic bin clustering and location classification, accurate waste status prediction, and priority-based route optimization, leading	Implements AI-driven forecasting using logistic regression and dynamic route optimization via a genetic algorithm but lacks real-time IoT sensing and digital-twin

				operations at lower cost.	open-source Philadelphia waste dataset.	to reduced traffic, fuel use, and pollution.	integration. The proposed project addresses these gaps through continuous IoT data, ML-based forecasting, and synchronized digital-twin routing for enhanced real-time operation
S24	Neha Gupta, 2025	<i>Comprehensive Mix-Method Approach for Efficient Waste Management using IoT-Driven Solutions</i> — IEEE	RQ2, RQ3, RQ5	Aimed to explore the potential of IoT technologies for intelligent, data-driven waste management by developing smart collection systems with predictive analytics and evaluating their operational, economic, and environmental benefits.	Mixed-method with quantitative sensor data, geospatial mapping & longitudinal tracking; applied ML (regression, time-series, SEM — specific models not specified) on real-time IoT data (ultrasonic, weight, temperature, RFID) using edge computing & predictive analytics.	Achieved up to 52% improvement in collection efficiency, 37% reduction in operational costs, and 66.7% decrease in carbon emissions, demonstrating significant savings and optimization gains in IoT-enabled waste management.	Strong in IoT sensing and AI-based predictive routing with measurable efficiency and environmental gains, but lacks digital-twin modeling and reported ML accuracy, marking a key gap addressed by the proposed project.
S25	Udhayakumar C., Nithin Nandha M., Pradeesh V. S., Praveen S., Rohith S. (2023)	<i>Smart Waste Management System for Metropolitan Areas</i> — IEEE	RQ2, RQ3	Aimed to develop an IoT-based smart waste management system for efficient household waste disposal, collection, and recycling in developing countries, addressing resource constraints and socioeconomic challenges.	Used IoT-enabled ultrasonic sensors with Arduino Uno for data collection, applying regression and classification models (specific algorithms not specified) with graph theory for route optimization and QR code technology for waste tracking.	Demonstrated an integrated framework combining real-time bin-level sensing, AI-based waste forecasting, and dynamic route optimization, resulting in improved time efficiency and reduced overflow incidents.	Implements IoT-based forecasting and dynamic routing using sensor data, aligning with the project's objectives. However, the forecasting model is not specified, and digital-twin synchronization is absent. The proposed project addresses these gaps through defined ML forecasting models,

S26	T. Thirupathieswaran, K. Ramalakshmi, and S. Karthik (2023)	<i>IoT Enabled Waste Management Optimization Framework (IWMOF)</i> — IEEE	RQ2, RQ3	To design an IoT-integrated and AI-driven waste management framework for efficient waste collection, adaptive scheduling, and route optimization.	IoT sensors for bin-level and truck data acquisition; Fuzzy Logic Control for interpreting sensor inputs; Deep Reinforcement Learning (DRL) for dynamic route and schedule optimization; implemented and simulated in MATLAB/Simulink.	Achieved 93.8% collection efficiency, 97.3% classification accuracy, and notable cost and emission reductions compared to GA, ANN, and PSO models. Demonstrated integration of IoT sensing, intelligent control, and adaptive routing.	continuous IoT sensing, and integrated digital-twin route optimization.
	Ferreira, J.A., Costa, M., Tereso, A., & Oliveira, J.A. (2015)	A Multi-Criteria Decision Support System for a Routing Problem in Waste Collection (Scopus)	RQ3	Aimed to develop a decision support system using route optimization and multi-criteria decision analysis to maximize waste collection efficiency under time, distance, and capacity constraints.	Used a Cellular Genetic Algorithm for route optimization and MCDA methods (SMART, ValueFn, AHP) within a CTOPTW model, tested on real-world waste management company data.	The DSS produced optimized route plans that improved scheduling flexibility and supported decision-makers in selecting efficient collection paths for a real Portuguese waste-management company.	Implements a waste generation prediction module and dynamic routing via a cellular genetic algorithm but lacks real-time IoT sensing and digital-twin integration. It does not involve IoT sensing, forecasting, or digital-twin modeling, which are addressed in the proposed project through continuous IoT data, ML forecasting, and synchronized digital-twin route optimization.
		Application of the Clarke–Wright Method to Improve	RQ3	Aimed to optimize waste collection using the Clarke–	Applied the Clarke–Wright Savings Algorithm and	Achieved a 50% reduction in travel distance, saving 1,469 L of fuel	Uses a static Clarke–Wright algorithm for route optimization, reducing fuel use and

S29	Mašek, J., Pálková, A., & Bulková, A. (2024)	the Sustainability of the Logistic Chain (<i>Scopus</i>)		Wright method to reduce costs and enhance sustainability and environmental performance in waste management.	Floyd–Warshall algorithm to solve a Capacitated Vehicle Routing Problem (CVRP) using custom software for route optimization.	annually and EUR 2,205 weekly, with four fewer vehicles required daily resulting in major financial and environmental benefits.	maintenance costs but lacking AI-based forecasting, real-time IoT sensing, and digital-twin integration. The proposed project addresses these gaps through continuous IoT data, ML forecasting, and dynamic digital-twin routing.
	Perumalla Dharan; V. M. Manikandan (2024)	<i>A Smart Waste Management System with Optimized Routing Method</i> — IEEE	RQ3	Aimed to develop an IoT- and ML-based smart waste management system to optimize collection routes, enhance public health and sustainability, and reduce costs and collection time.	Used Arduino MKR WiFi 1010 with ultrasonic sensors for IoT data collection and applied ML algorithms (not specified) with the pyvrp module for route optimization based on distance, capacity, and fuel use, with matplotlib for visualization.	Demonstrates real-time bin monitoring and a working VRP-based route planner; presents workflow and prototype screens. No city-scale trials or quantified KPIs.	Introduces smart bins with real-time IoT monitoring and ML-based route optimization but lacks continuous IoT sensing, ML forecasting, and digital-twin integration.
S30	Banupriya, V.; Renugadevi, P.; Arul Rixy Catherine, T.; Unnamalai, C.; Gopika Shri, M. (2025)	<i>AI-Driven Bio-Waste Classification and Bin Management System Using Sensor Fusion and IoT</i> — IEEE	RQ3	Aimed to develop an AI-driven bio-waste classification and IoT-based bin management system using sensor fusion to improve segregation, optimize routes, and support sustainability and circular economy goals.	Used AI with sensor fusion and IoT technology, applying CNNs, threshold, and hazard detection algorithms with predictive analytics on data from infrared, ultrasonic, gas, image, and weight sensors, supported by cloud-based analytics.	Achieved 95% classification accuracy and 80% reduction in manual effort, reducing landfill use by 40%. Demonstrated high hazard detection precision, lower fuel consumption, and reduced operational costs through route optimization.	Uses predictive analytics conceptually for smart decision-making but does not implement or evaluate any forecasting model. Routing is adaptive with real-time IoT sensing, but digital-twin synchronization and explicit fill-level forecasting are absent.
S31	Cárdenas-León, I., Koeva, M., Davey, C., & Nourian, P.	(2024). <i>Solid Waste in the Virtual World: A Digital Twinning Approach for Waste</i>	RQ4	Proposed a digital-twin framework for waste management in South Africa using	Used Epicollect5 for crowd-sourced data, processed via PyEpiCollect and GIS	Mapped 1,270 containers and 820 littering sites, revealing uneven	Conceptual digital twin only; lacks AI forecasting, real-time IoT sensing, and

		<i>Collection Planning. In Lecture Notes in Geoinformation and Cartography (Springer).</i>		citizen participation to map containers and reduce illegal dumping.	(2D/3D) with a conceptual digital twin and referenced CVRP for optimization.	distribution and showing the value of citizen engagement for improved planning.	dynamic routing—CVRP mentioned but not implemented.
S32	Cárdenas-León, M., Llorente-Álvarez, A., González-Enríquez, C., & Caballero-Sánchez, R. (2024)	<i>Urban Digital Twin-Based Solution Using Geospatial Information for Solid Waste Management (Scopus)</i>	RQ4	Developed an Urban Digital Twin for solid waste management with waste generation simulations, stakeholder input, and route optimization to improve efficiency and resource use.	Applied Tabu search for dynamic route optimization via CVRP solver (ESRI, 2023); used GIS data (CoT GIS, OSM, NGI) and stakeholder evaluation through AHP and Gemini Principles.	Reduced fuel use, costs, and emissions (~1.9 M ZAR savings); optimized routes (236 km avg.); strong stakeholder approval for usability and decision support.	Implements digital-twin-based dynamic routing but lacks AI-driven fill-level forecasting and real-time IoT sensing for adaptive synchronization.
S33	Velásquez W., Dillon-Jimenez J., Medina-Moncayo D., Alvarez-Alvarado M. (2024)	<i>Optimizing Waste Collection Routes with QPSO-LR Algorithm and IoT Sensors — IEEE</i>	RQ5	Developed a web application using the Lorentz Quantum Particle Swarm Optimization (QPSO-LR) algorithm to optimize waste collection routes, reducing time, distance, and environmental impact.	Applied QPSO-LR with simulated IoT sensor data for container fill levels; used Google Maps for geolocation, AWS for cloud hosting, and a 66-point ESPOL campus dataset.	Significantly reduced collection distance and time, improving operational efficiency, fuel use, and cost savings while lowering emissions.	Implements dynamic route optimization using QPSO-LR with simulated IoT sensing but lacks AI-based forecasting and digital-twin synchronization for real-time adaptive management.
S34	Kuzhin, M., Dzhonov, S., & Aliyev, S. (2024)	<i>Optimizing Waste Management through IoT and Analytics: A Case Study Using the Waste Management Optimization Test (Scopus)</i>	RQ5	Investigated the integration of IoT and analytics to improve waste management efficiency, reduce costs, and enhance sustainability through real-time	Deployed IoT sensors on bins for real-time fill-level data and applied advanced analytics, predictive modeling, and machine learning (specific models not specified) for optimization and recycling insights.	Reduced overfilled bins by 20%, trip distance by 25%, fuel use by 10%, and costs by 12%, while improving recyclables collected by 15%. Demonstrated the environmental and	Implements real-time IoT sensing and dynamic optimization using analytics but lacks explicit forecasting models, AI-driven prediction, and digital-twin integration.

S35				data collection and route optimization.		economic benefits of data-driven waste management.	
	Belhiah, M.; El Aboudi, M. — 2025.	An IoT-Based Sensor Mesh Network Architecture for Waste Management in Smart Cities (Scopus)	RQ5	Designed an IoT mesh-network architecture for smart bin management to enable real-time monitoring, reduce infrastructure costs, and optimize waste collection in areas without LoRa coverage.	Implemented mesh topology with ultrasonic sensors, custom firmware, GPS and RTC modules, dynamic routing, and a three-layer IoT architecture for real-time data transfer.	Reduced hardware costs from \$16,600 to \$700 and connectivity costs from \$23,520 to \$1,680, improved data reliability (0.5% loss), lowered power use by 81%, and enhanced route optimization.	Demonstrates real-time IoT sensing and dynamic routing but lacks AI-based forecasting and digital-twin synchronization.
	Al Yarubi, K. S.; Khairy, S. O. F.; Hossain, S. M. E.; Hayder, G. (2025)	Internet of Things-Driven Waste Management: Paving the Way for Sustainable Smart Cities(Scopus)	RQ5	Developed an IoT- and AI-based waste management system in Al-Duqm to enhance efficiency, cut costs, and support sustainability.	Integrated ultrasonic and infrared sensors with GPS/GSM and cloud-based monitoring; proposed an AI module (specific model not specified) for route optimization within the IoT architecture.	Improved efficiency by 41.5%, reduced fuel use by 28%, costs by 15%, and raised collection rate from 65% to 92%.	Implements IoT monitoring but only proposes—not implements—AI routing; lacks forecasting and digital-twin functionality.
	Martikkala, A., Mayanti, B., Helo, P., Lobov, A., & Flores Ituarte, I. (2023)	Smart textile waste collection system – Dynamic route optimization with IoT (Scopus)	RQ5	Developed a low-cost IoT-based textile waste collection system with dynamic route optimization to reduce costs, distance, and emissions using real-world data.	Used Heltec CubeCell LoRa sensors with InfluxDB and Firestore for data storage, Grafana for visualization, and Open-Door Logistics for dynamic route optimization under the DIKW framework.	Reduced costs by 7.4%, CO ₂ emissions by 10.2%, and time by 7.3%; improved collection efficiency and reduced overflows, validating IoT-enabled dynamic routing.	Implements real-time IoT sensing and dynamic routing using open-source tools but lacks AI-based forecasting and digital-twin integration.
S38	Mustapha Aouache, Zaid Bousmina, Ilyes Keskass, Zakarya	<i>Practical IoT-Enabled Monitoring Platform for Solid</i>	RQ5	Developed an IoT-based system (I-Bin) for real-time	Used HC-SR04 ultrasonic and DHT-11 sensors with a	Improved route efficiency and reduced travel time,	Demonstrates IoT sensing and dynamic routing but lacks AI

S39	Guettatfi, Abdelfetah Hentout, Ali Mansoul (2021)	<i>Waste Collection: A Case Study for the Municipality of Baba Hassen, Algiers, Algeria</i> — IEEE		bin monitoring and route optimization to enhance waste collection in Algiers.	PIC18F2550 microcontroller, GSM/GPRS module, NoSQL database, and A* algorithm for dynamic routing.	fuel use, and labor costs through real-time bin data and adaptive scheduling.	forecasting and digital-twin integration. es.
	Ahmad, S., Imran, F., Jamil, F., Iqbal, N., & Kim, D. (2020)	Optimal Route Recommendation for Waste Carrier Vehicles for Efficient Waste Collection: A Step Forward Towards Sustainable Cities (Scopus)	RQ5	Proposed an intelligent system to predict waste levels and optimize routes for carrier vehicles in Jeju Island.	Used Support Vector Regression (SVR) for waste prediction and PSO, GA, and BAT algorithms for multi-objective route optimization.	Reduced travel distance and fuel use; PSO achieved best cost results, BAT fastest convergence; validated predictive route optimization.	Implements AI forecasting and dynamic routing but lacks IoT-based sensing and digital-twin integration.
S40	Alya Nasir Alnuaimi, Ridvan Aydin, Sameh Al-Shihabi (2024)	<i>Optimizing Municipal Solid Waste Collection Using Smart Bins: A Case Study in the United Arab Emirates</i> — IEEE	RQ5	Evaluated smart bins and route optimization in the UAE to enhance waste collection efficiency and cut fuel, cost, and emissions.	Used IoT bin data in an AnyLogic simulation with Travelling Salesman (TSP) and Nearest Neighbor (NN) algorithms for route optimization.	Reduced route distance from 57 km to 18.6 km, lowering fuel, cost, and CO ₂ ; TSP provided more consistent optimization.	Uses IoT sensing and static route optimization but lacks forecasting, real-time adaptation, and digital-twin integration.
S41	A. Sadikin, M. R. Rachman, F. R. Rahardja, R. P. Sari (2025)	<i>TrashTrack: A Smart Waste Monitoring System with Constraint-Based Greedy Algorithm for Optimized Collection Routes</i> — IEEE	RQ5	Developed TrashTrack, an IoT-based system using a constraint-based greedy algorithm for efficient waste collection routing.	Used ultrasonic and infrared sensors with ESP32, GPS, Firebase cloud, and Android app for real-time bin tracking and route optimization.	Improved routing efficiency, reduced redundant trips, and achieved high usability (≈92% effectiveness and usefulness).	Demonstrates IoT sensing and dynamic routing but lacks AI forecasting and digital-twin integration.